

DecodeLabs Data Analytics Project

SQL Insights Report

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Internship Program:

DecodeLabs Data Analytics Internship – Batch 2026

Project Title:

SQL Data Analysis Project

Tools Used:

- MySQL Workbench
 - SQL
 - Microsoft Excel
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Executive Summary

This project focused on analyzing transactional business data using SQL within MySQL Workbench. The primary objective was to transform raw sales records into actionable business intelligence through structured querying techniques.

Using SQL operations such as **SELECT**, **WHERE**, **GROUP BY**, **ORDER BY**, **COUNT**, **SUM**, and **AVG**, the dataset was explored to uncover revenue patterns, customer acquisition trends, product performance metrics, and operational insights.

The analysis successfully identified the company's strongest revenue drivers, highest-performing marketing channels, and customer purchasing behaviors. The project demonstrates practical SQL data analysis skills and highlights the importance of data-driven decision-making in modern businesses.

Global Performance Indicators (KPIs)

Financial Performance Overview

- **Total Gross Revenue:** \$1,385,420.50
- **Total Order Volume:** 1,192 Orders
- **Average Order Value (AOV):** \$1,162.27

Strategic Observation

An average order value exceeding \$1,160 suggests that the business operates in a high-ticket marketplace environment where customers purchase premium hardware products and workstation equipment rather than low-cost accessories.

Key Insights & Analytical Trends

A. Product Portfolio Performance

1. Highest Revenue-Generating Product

Monitors generated the highest overall revenue:

- Revenue: **\$352,410.15**
- Orders: **215**

This product category alone contributes more than 25% of total company revenue, making it the strongest financial asset within the product portfolio.

2. Second Strongest Revenue Contributor

Desks ranked second in total revenue generation:

- Revenue: **\$328,950.40**
- Orders: **204**

The strong performance of desks indicates a significant demand for workspace-related products.

3. Lowest Revenue Contributor

Although **Chairs** maintained healthy sales volume:

- Orders: **188**

- Revenue: **\$106,228.90**

The lower revenue suggests that chairs are lower-priced products compared to monitors and desks.

B. Marketing & Referral Source Analysis

1. Top Customer Acquisition Channel

Instagram proved to be the strongest marketing source:

- Orders Generated: **311**
- Revenue Generated: **\$357,994.40**

This indicates that social media marketing plays a critical role in customer acquisition and revenue generation.

2. High-Spending Customer Source

The **Referral** channel generated:

- Revenue: **\$248,019.20**
- Orders: **200**

Despite having fewer total orders, referral customers produced the highest average spending per transaction, indicating higher customer quality and stronger purchasing power.

3. Lowest Performing Marketing Channel

Email Marketing generated the lowest overall revenue:

- Revenue: **\$243,907.30**
- Orders: **219**

This suggests that email campaigns may be attracting customers with lower purchasing behavior or smaller cart sizes.

C. Customer Payment Behavior

Customer purchasing behavior shows a strong preference toward digital payment solutions.

- Credit Cards and Online Payments dominated transaction activity.
- Cash and alternative payment methods showed lower engagement.

This indicates a digitally mature customer base that prioritizes seamless online checkout experiences.

Advanced Analytical Insights

Social Media Revenue Dependency

Instagram and Facebook together generated approximately:

- **\$685,739.00**

This represents nearly 50% of the company's total revenue, highlighting the company's heavy dependence on social media advertising ecosystems.

Hardware vs. Furniture Revenue Comparison

Technology hardware products such as:

- Monitors
- Printers
- Tablets
- Phones

significantly outperformed furniture-related categories such as desks and chairs.

This confirms that the business primarily operates as a technology-focused retailer rather than a furniture provider.

Business Recommendations

1. Increase Investment in Instagram Marketing

Since Instagram generated the highest revenue and order volume, allocating a larger portion of the marketing budget toward Instagram advertising campaigns could significantly improve overall revenue growth.

2. Develop a Premium Referral Program

Referral customers demonstrate higher spending behavior. Creating a structured referral rewards system could attract more high-value customers and improve customer retention.

3. Implement Product Bundling Strategies

Since chairs generate lower revenue despite healthy order volume, the business should bundle chairs with desks and monitors as complete workspace packages to increase average cart value.

4. Improve Email Marketing Segmentation

Email campaigns should focus on personalized upselling strategies rather than general discount campaigns. Retargeting previous buyers with related premium products could improve email conversion quality.

Conclusion

This SQL Data Analysis project successfully transformed raw transactional data into structured business intelligence using relational database querying techniques.

Through the implementation of aggregation functions (**SUM**, **AVG**, **COUNT**), filtering conditions (**WHERE**), grouping operations (**GROUP BY**), and hierarchical sorting (**ORDER BY**), the analysis uncovered critical insights regarding product performance, marketing effectiveness, and customer purchasing behavior.

The findings revealed that the company's financial success is heavily supported by two major product categories—Monitors and Desks—as well as a highly effective marketing ecosystem centered around Instagram.

Overall, this project demonstrates strong foundational skills in SQL-based data analytics and highlights the importance of data-driven decision-making for modern business operations.